

claude-windows / 068560d3-1ca2-4387-981a-d2d6bca55414

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\068560d3-1ca2-4387-981a-d2d6bca55414\subagents\workflows\wf\_47057438-9c9\agent-a69b45904982be16e.jsonl`
- SHA-256: `e6232ce85bc73dfc1bb42ab842609c359134e7e4e58d1ce823f2fa7f03fd6790`
- Source modified: `2026-06-16T03:00:47+00:00`
- Imported at: `2026-07-05T16:48:28+00:00`
- project: `wf\_47057438-9c9`
- session_id: `068560d3-1ca2-4387-981a-d2d6bca55414`

Transcript

1. user (2026-06-16T02:57:43.898Z)

You are a SKEPTICAL second lawyer. Try to REFUTE or materially QUALIFY the bottom line below using authoritative sources. Look for the strongest counter-authority, jurisdictional exceptions, and ways an over-confident reading would get the client in trouble. Default to 'qualified' or 'refuted' if there is genuine doubt.

METHOD: Use WebSearch and WebFetch to find and cite AUTHORITATIVE sources (statutes, case law, official regulator guidance, reputable legal commentary). Load them first if needed via ToolSearch with query 'select:WebSearch,WebFetch'. Cite real URLs. If you cannot fetch a source, mark that claim's confidence 'low' and set webUsed=false. Do NOT fabricate citations ? an honest 'unverified' beats an invented case name or URL.

QUESTION KEY: privacy-biometrics

BOTTOM LINE TO ATTACK: Triage by what the stream is ABOUT and whether it can single out a person. (1) SHUTTLE coordinates describe an inanimate object ? not personal data, and safe to publish, UNLESS you join them to data that singles out a specific player. (2) PERSON BOUNDING BOXES (detect/locate/count, or estimate age/gender) are ordinary personal data when they single out a human, but are NOT Art.9 biometric data as long as the purpose is to detect/categorize, not uniquely identify ? publishing coordinates stripped of any identity link is low-risk; publishing the cropped images is identifying personal data. (3) POSE KEYPOINTS / GAIT are the danger zone: gait is a behavioural biometric that re-identifies people even with faces blurred, and skeleton 'anonymization' provably leaves gait identity intact ? so raw pose/gait tracks are NOT reliably anonymous, are personal data when usable to single out a player, and become Art.9 special-category (needing explicit consent) the moment they are used to uniquely identify. Practical publish/no-publish: shuttle tracks = publish freely; bbox/pose COORDINATES detached from any identity and used only for rally detection = generally OK with normal personal-data hygiene; named player + gait/pose signature, or any face crop = do NOT publish without an explicit-consent Art.9 basis. MINORS flip everything: under India's DPDP (child = under 18) you need VERIFIABLE parental consent (Rule 10 ? real ID/age verification, not a checkbox) before processing ANY child's data, you may not cause detrimental effects, and s.9(3) flatly BANS tracking/behavioural monitoring of children ? which a persistent pose/gait signature arguably is. The cleanest path for footage containing minors is to exclude them from any persisted derived stream beyond inanimate shuttle data, or obtain verified parental consent and keep only de-identified object/coordinate data. This matches the project's existing guardrails: exclude minors from training, treat gait/biometrics as strictly future (GDPR Art.9), and keep a generic schema.

SUPPORTING FINDINGS (JSON): [{"claim":"GDPR only protects information about an identified or identifiable NATURAL PERSON ? data purely about an inanimate object is not personal data on its face. BUT 'relating to' an object can still make data personal if it singles out / is used about a person (Art.29 WP136 content/purpose/result test). So shuttle (shuttlecock) coordinates are about an object and are not personal data UNLESS they are used to single out or identify a specific player.", "support":"Art.4(1) GDPR defines personal data as 'any information relating to an identified or identifiable natural person'. WP136 (Opinion 4/2007): data 'relate to' a person via a content OR purpose OR result element, and the CJEU confirms 'data that relate to an object can identify a person.' Shuttle trajectories describe the object's position, not a person's body

? no content/purpose/result link to a specific individual on their own.", "sourceName": "gdpr-info.eu ? Personal Data; Oxford IDPL / Bird & Bird on WP136 'relates to' test", "sourceUrl": "https://gdpr-info.eu/issues/personal-data/", "confidence": "high"}, {"claim": "Video footage and ANY derived stream that lets you single out a specific human is personal data, even with no name attached ? the bar for 'anonymous' (escaping GDPR entirely) is very high. Recital 26: identifiability is judged by ALL means 'reasonably likely to be used' (singling out, cost, time, available + foreseeable technology). Pseudonymized data (re-linkable with extra info) is still personal data.", "support": "Recital 26: principles 'should not apply to anonymous information ... rendered anonymous in such a manner that the data subject is not or no longer identifiable'; identifiability turns on 'all the means reasonably likely to be used, such as singling out.' Pseudonymized data that 'could be attributed to a natural person by the use of additional information' remains personal data.", "sourceName": "gdpr-info.eu ? Recital 26 (Not Applicable to Anonymous Data)", "sourceUrl": "https://gdpr-info.eu/recitals/no-26/", "confidence": "high"}, {"claim": "Person BOUNDING BOXES (detect/locate/count people, or categorize age/gender) are ordinary personal data, NOT Art.9 special-category biometric data ? provided the purpose is to distinguish a category of people, not to uniquely identify a named individual. EDPB: 'When the purpose of the processing is ... to distinguish one category of people from another but not to uniquely identify anyone, the processing does not fall under Article 9.'", "support": "EDPB Guidelines 3/2019 on processing of personal data through video devices, as summarized by Fox Rothschild: video that categorizes without unique identification does not trigger Art.9; Art.9 applies only 'if the controller stores biometric data in order to uniquely identify a person.' Counting/age-gender-estimation examples fall outside Art.9.", "sourceName": "Fox Rothschild ? Video Surveillance Under GDPR: EDPB Issues Final Guidance (summarizing EDPB Guidelines 3/2019)", "sourceUrl": "https://www.foxrothschild.com/publications/video-surveillance-under-gdpr-edpb-issues-final-guidance", "confidence": "high"}, {"claim": "Biometric data becomes Art.9 SPECIAL-CATEGORY only when processed 'for the purpose of uniquely identifying a natural person.' Behavioural traits ? gait, typing rhythm, and by extension distinctive movement signatures ? are within Art.4(14) biometric data, and become special-category once used for unique identification. The trigger is PURPOSE + technical processing that enables matching, not the raw modality.", "support": "Art.9(1) GDPR prohibits processing of 'biometric data for the purpose of uniquely identifying a natural person.' Art.4(14) defines biometric data as data from specific technical processing of 'physical, physiological or behavioural characteristics ... which allow or confirm the unique identification.' ICO/commentary: not all biometric data is special-category ? only if used to uniquely identify; gait is named as a behavioural biometric.", "sourceName": "gdpr-info.eu ? Art.9 GDPR; ICO biometric-recognition key concepts", "sourceUrl": "https://gdpr-info.eu/art-9-gdpr/", "confidence": "high"}, {"claim": "POSE KEYPOINTS / GAIT are the genuinely risky derived stream: peer-reviewed work shows gait is a biometric identifier that re-identifies people even with faces obscured, and that pose/skeleton 'anonymization' removes appearance but LEAVES GAIT IDENTITY LARGELY INTACT (80?95% gait-recognition accuracy on anonymized data). So publishing raw skeleton/pose tracks is NOT reliably anonymous, and if used/usable to single out a player it can be personal data ? and special-category biometric data if the purpose is unique identification.", "support": "Weiß et al., 'Privacy Beyond the Face: Assessing Gait Privacy Through Realistic Anonymization in Industrial Monitoring,' Sensors (2025): 'human pose and gait patterns represent biometric identifiers capable of revealing identity even when the face is obscured'; 'anonymization removes appearance but leaves gait identity largely intact, indicating that pose-driven anonymization is insufficient for privacy protection.'", "sourceName": "Weiß et al., Sensors 2025 (PMC) ? Privacy Beyond the Face", "sourceUrl": "https://pmc.ncbi.nlm.nih.gov/articles/PMC12788357/", "confidence": "high"}, {"claim": "MINORS change the analysis on both axes. EU: Art.8 makes a child's consent to information-society-services processing lawful only at 16+ (Member States may lower to a floor of 13), otherwise parental consent + reasonable verification is required; and any biometric/special-category processing of a child stacks Art.9 on top. India DPDP s.9 is stricter: a 'child' is anyone under 18 (s.2(f)), and BEFORE processing any child's personal data the fiduciary must obtain VERIFIABLE parental/guardian consent, must not cause detrimental effect, and is flatly barred from tracking, behavioural monitoring, or targeted advertising of children.", "support": "GDPR Art.8(1): processing of a child's data for ISS is lawful where the child is 16+, else needs parental authorization; Member States may set a lower age 'not below 13'; controller must make 'reasonable efforts to verify.' DPDP s.2(f): child = under 18. DPDP s.9(1) requires 'verifiable consent of the parent ... or the lawful guardian'; s.9(2) bars detrimental processing; s.9(3) bars 'tracking or behavioural monitoring of children or targeted advertising directed at children.'", "sourceName": "gdpr-info.eu ? Art.8 GDPR; dpdpa.com ? DPDP Act 2023 Section 9", "sourceUrl": "https://gdpr-info.eu/art-8-gdpr/", "confidence": "high"}, {"claim": "India's DPDP Rules 2025 (Rule 10) operationalize 'verifiable' parental consent: the fiduciary must use technical/organizational

measures to confirm the consent is an identifiable adult (18+) and the parent ? either via identity/age details it already reliably holds, or details voluntarily provided (incl. a virtual token issued by an authorized entity such as a government agency or DigiLocker provider). A bare 'I am over 18' checkbox is not sufficient.", "support": "DPDP Rules 2025, Rule 10(1): fiduciary must 'adopt appropriate technical and organisational measures to ensure that verifiable consent of the parent is obtained' and verify via 'reliable details of identity and age ... available with the Data Fiduciary' or details/virtual-token from the individual; 'adult' = completed 18 years.", "sourceName": "dpdpa.com ? DPDP Rules 2025, Rule 10", "sourceUrl": "https://www.dpdpa.com/dpdparules/rule10.html", "confidence": "medium"}, {"claim": "Behavioural biometric identification (which gait/pose enables) is widely treated as Art.9 special-category once used to identify, and regulators are increasingly strict on such 'behavioural' processing ? reinforcing that a movement-signature stream tied to an identified athlete is high-risk to publish without an Art.9 lawful basis (typically explicit consent), and is essentially off-limits for minors under DPDP s.9's behavioural-monitoring ban.", "support": "Commentary on Art.9 (GDPR Local; ICO) confirms gait/typing-rhythm are behavioural biometrics that become special-category when processed to uniquely identify; DPDP s.9(3) independently prohibits behavioural monitoring of children regardless of identification purpose.", "sourceName": "GDPR Local ? Biometric Data GDPR Compliance; ICO biometric guidance", "sourceUrl": "https://gdprlocal.com/biometric-data-gdpr-compliance-made-simple/", "confidence": "medium"}]}

2. assistant (2026-06-16T02:57:47.824Z)

I'll act as a skeptical second lawyer, hunting for counter-authority and ways an over-confident reading of this bottom line could get the client in trouble. Let me load the web tools first.

3. assistant (2026-06-16T02:57:54.411Z)

I'll attack this bottom line systematically. Let me start by probing the weakest joints: (1) whether shuttle/object coordinates are truly "safe to publish," (2) whether bbox/pose coordinates are reliably outside Art.9, (3) the DPDP s.9(3) behavioural-monitoring claim and Rule 10 verification, and (4) whether "publishing coordinates" can be cleanly separated from identifiability when the footage exists.

Let me run parallel searches on the strongest counter-authorities.

4. user (2026-06-16T02:58:03.532Z)

Web search results for query: "EDPB Guidelines 3/2019 video devices biometric data unique identification Article 9 categorization"

Links: [{"title": "Guidelines 3/2019 on processing of personal data ... - EDPB", "url": "https://www.edpb.europa.eu/sites/default/files/files/file1/edpb_guidelines_201903_video_devices.pdf"}, {"title": "GDPR Update - EDPB Video Surveillance Guidelines - Data Protection - Netherlands", "url": "https://www.mondaq.com/data-protection/842286/gdpr-update-edpb-video-surveillance-guidelines"}, {"title": "When is voice (a special category of) personal data under GDPR?", "url": "https://www.linkedin.com/pulse/when-voice-special-category-personal-data-under-gdpr-janvier-parewyck"}, {"title": "Guidelines 3/2019 on processing of personal data through ...", "url": "https://www.edpb.europa.eu/sites/default/files/files/file1/edpb_guidelines_201903_video_devices_en_0.pdf"}, {"title": "EDPB Adopts Guidelines on Data Processing Through Video Devices", "url": "https://www.hunton.com/privacy-and-information-security-law/edpb-adopts-guidelines-on-data-processing-through-video-devices"}, {"title": "EDPB guidelines on video devices ? CIPP Training.com", "url": "https://cipptraining.com/edpb-guidelines-on-processing-through-video-devices/"}, {"title": "edpb guidelines 201903 video devices en", "url": "https://www.edpb.europa.eu/sites/default/files/files/file1/edpb_guidelines_201903_video_devices_en.pdf"}]}

Based on the search results, here's what the EDPB Guidelines 3/2019 states regarding video devices, biometric data, unique identification, and Article 9 categorization:

****Key Requirement for Article 9 Classification:****

In order for it to be considered as processing of special categories of personal data (Article 9), biometric data must be processed "for the purpose of uniquely identifying a natural person".

****Video Footage vs. Biometric Data:****

Video footage of an individual cannot, however, in itself be considered as biometric data under Article 9 GDPR if it has not been specifically and technically processed to contribute to the identification of an individual (i.e. for facial recognition).

****Examples of Article 9 Processing:****

The Guidelines provide specific examples: A hospital installing a video camera in order to monitor a patient's health condition would be considered as processing of special categories of personal data (Article 9). Additionally, political opinions could for example be deduced from images showing identifiable data subjects taking part in an event, engaging in a strike, etc., which would fall under Article 9.

****Risk Considerations:****

Processing of special categories of data and, in particular, biometric data (e.g., when using facial recognition technology) entail heightened risks for individuals and therefore require an increased and continued vigilance with regard to certain obligations: e.g., a high level of security and a data protection impact assessment, where necessary.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

5. user (2026-06-16T02:58:04.387Z)

Web search results for query: "DPDP Act section 9(3) tracking behavioural monitoring children exemption rules 2025 verifiable consent"

Links: [{"title": "Prohibition of Behavioral Tracking and Targeted Advertising for Children Under the DPDP Act, 2023", "url": "https://cyberpeace.org/resources/blogs/prohibition-of-behavioral-tracking-and-targeted-advertising-for-children-under-the-dpdp-act-2023"}, {"title": "Rethinking Regulation of Behavioural Monitoring under DPDP Act", "url": "https://www.medianama.com/2025/03/223-rethinking-behavioural-monitoring-targeted-ads-regulation-dpdp-act/"}, {"title": "Section 9 in THE DIGITAL PERSONAL DATA PROTECTION ACT, 2023", "url": "https://indiankanoon.org/doc/98869575/"}, {"title": "Children's Data ? Section 9 of the DPDP Act - DPDP India", "url": "https://www.dpdpindia.in/dpdp-children.html"}, {"title": "Expert Data Privacy and Security Blog - Insights from Tsaaro", "url": "https://tsaaro.com/blogs/privacy-for-minors-in-dpdpa/"}, {"title": "Safeguarding Minors Online: Understanding Parental Consent Obligations and Behavioural Monitoring Restrictions Under the DPDP and DPDP Rules", "url": "https://tsaaro.com/blogs/safeguarding-minors-online-understanding-parental-consent-obligations-and-behavioural-monitoring-restrictions-under-the-dpdpa-and-dpdp-rules"}, {"title": "Digital Personal Data Protection Act, 2023 DPDP SECTION 9 WITH INTERPRETATION", "url": "https://www.dpdpa.com/dpdpa2023/chapter-2/section9.html"}, {"title": "Rule 12 of Digital Personal Data Protection Act, 2023 DPDP Rules 2025", "url": "https://www.dpdpa.com/dpdparules/rule12.html"}, {"title": "Targeted Advertising Ban DPDP & Behavioral Monitoring of Children India | WatchDog Security", "url": "https://watchdogsecurity.io/dpdp/prohibition-on-behavioral-tracking"}, {"title": "How to Handle Children's Data Under India's DPDP Act", "url": "https://hypertrust.one/blog/childrens-data-under-dpdp/"}]

Based on the web search results, here's information about DPDP Act Section 9(3) regarding tracking and behavioral monitoring of children:

Section 9(3) - Core Prohibition

Section 9 sub-section (3) specifies that a Data Fiduciary shall not undertake tracking or behavioural monitoring of children or targeted advertising directed at children.

Exemption Framework

Section 9(4) empowers the Central Government to notify, by Rules, certain Data Fiduciaries or classes of Data Fiduciaries for whom Section 9(1) (parental consent) and Section 9(3) (prohibitions) will not apply, or will apply with modifications.

Potential categories being considered for exemption: Educational technology platforms, healthcare platforms for minors, government welfare schemes serving children, parental monitoring/safety apps.

DPDP Rules 2025 - Rule 12 Exemptions

Rule 12 of the DPDP Rules, read with the Fourth Schedule exempts certain classes of Data Fiduciaries and processing activities from complying with the requirement of verifiable parental consent under Section 9(1) and the prohibition of behavioural tracking or targeted advertisements directed towards children under Section 9(3) of the DPDP Act.

Certain fiduciaries such as hospitals/healthcare establishments, schools/educational institutions, day-care/creches and child-transport services may process child data with limited relief from the verifiable-consent requirement, subject to conditions.

Verifiable Consent Requirements

Rule 10 of the DPDP Rules imposes an obligation on the data fiduciary to adopt appropriate technical and organisational measures to ensure that verifiable consent of the parent is obtained before any personal data of a child is processed.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-06-16T02:58:10.598Z)

Two important counter-points are already emerging. Let me dig into the strongest refutations: (a) the EDPB's actual position on whether mere video footage of identifiable people IS personal data (and the "singling out" risk that undercuts "publish coordinates freely"); (b) the DPDP Rule 12 / Fourth Schedule exemptions which materially qualify the "s.9(3) flatly BANS" claim; and (c) whether shuttle coordinates joined to broadcast/known-match context can single out players. Let me fetch the primary sources.

7. user (2026-06-16T02:58:16.750Z)

Rule 12 DPDP Rules 2025 ? Exemptions for Child Data Processing

Verbatim Text of Rule 12

"(1) The provisions of sub-sections (1) and (3) of section 9 of the Act shall not be applicable to processing of personal data of a child by such class of Data Fiduciaries as are specified in Part A of Fourth Schedule..."

"(2) The provisions of sub-sections (1) and (3) of section 9 of the Act shall not be applicable to processing of personal data of a child for such purposes as are specified in Part B of Fourth Schedule..."

Exempted Classes & Purposes

The webpage references **Part A and Part B of the Fourth Schedule** but does not reproduce their full text. The legal interpretation section identifies these exempted entity categories:

- **Healthcare entities** (hospitals/medical establishments)
- **Educational institutions** (schools)
- **Child care services** (day-care/creches)
- **Child transport services**

Purpose-based exemptions (Part B) include:

- Emergency medical care
- Child safety monitoring
- Statutory compliance

- Research/archival activities (narrowly defined)

Applicable Conditions

All exemptions require: necessity and proportionality, data minimisation, documentation with Data Protection Impact Assessments, technical/organisational safeguards, and transparency notices where feasible. Processing must remain "strictly necessary and proportionate to the stated institution/purpose."

8. user (2026-06-16T02:58:19.935Z)

Section 9(3) Ban: Scope and Definitions

Based on the provided content, here's what the article reveals:

Stated Scope

The article indicates that **Section 9(3) prohibits "tracking or behavioural monitoring of children or targeted advertising directed at children."** However, the document does not provide a formal legal definition of "tracking" itself.

Key Ambiguity

The article does not clarify whether the ban applies to:

- All behavioral monitoring activities, or
- Only monitoring conducted specifically for targeted advertising/commercial profiling

Exemptions

Importantly, the document notes that **"section 9 sub-section (5) further provides room for exemption from this prohibition by empowering the Central Government which may notify exemption to specific data fiduciaries or data processors."**

Critical Gap

The article makes no reference to the DPDP Rules 2025 or Rule 12/Fourth Schedule. The text states the rules are "yet to be announced or released," suggesting this article predates their publication (published July 11, 2024).

Bottom line: This source does not provide detailed definitions, scope clarification, or information about Rule 12 exemptions. For comprehensive answers about those elements, you would need to consult the actual DPDP Rules 2025.

9. assistant (2026-06-16T02:58:25.656Z)

The DPDP Rule 12 exemptions are real and material ? but importantly, an amateur badminton highlight indexer is NOT a hospital/school/childcare/transport entity, so those class exemptions likely don't help the client. Let me now attack the two most load-bearing GDPR claims: (1) whether the EDPB treats video of identifiable people as personal data even when "coordinates are stripped" (the "publish coordinates freely" claim), and (2) the strongest counter-authority on whether pseudonymized/singling-out data escapes GDPR. Let me also probe the German "skeleton/pose can be Art.9" angle and the medianama critique on s.9(3) scope.

10. user (2026-06-16T02:58:28.981Z)

The server returned HTTP 403 Forbidden.

The response body was not retrieved. If this URL requires authentication, use an authenticated tool (e.g. `gh` for GitHub, or an MCP-provided fetch tool) instead of WebFetch.

11. user (2026-06-16T02:58:38.933Z)

Web search results for query: "CJEU Breyer Nowak singling out pseudonymisation "means reasonably likely" identifiability personal data threshold relative absolute"

Links: [{"title": "Pseudonymized data after EDPS v SRB | Clifford Chance", "url": "https://www.cliffordchance.com/insights/resources/blogs/talking-tech/en/articles/2025/09/pseudonymized-data-after-edps-v-srb.html"}, {"title": "Pseudonymised Data: CJEU provides clarification on the concept of Personal Data - Burges Salmon", "url": "https://www.burges-salmon.com/articles/10215dc/pseudonymised-data-cjeu-provides-clarification-on-the-concept-of-personal-data/"}, {"title": "It's nothing personal - Reassessing pseudonymised data and AI after EDPS v SRB", "url": "https://www.lewissilkin.com/insights/2025/12/18/its-nothing-personal-reassessing-pseudonymised-data-and-ai-after-edps-v-srb-102ly5u"}, {"title": "CJEU clarifies concepts of personal and pseudonymised data: implications for data sharing | Osborne Clarke", "url": "https://www.osborneclarke.com/insights/cjeu-clarifies-concepts-personal-and-pseudonymised-data-implications-data-sharing"}, {"title": "Pseudonymized Data: Key Takeaways from CJEU Decision | Knowledge | Fasken", "url": "https://www.fasken.com/en/knowledge/2025/12/pseudonymized-data-key-takeaways-from-cjeu-decision"}, {"title": "Personal Data or Not? The CJEU's (updated) understanding of anonymisation | Freshfields", "url": "https://www.freshfields.com/en/our-thinking/blogs/technology-quotient/personal-data-or-not-the-cjeus-updated-understanding-of-anonymisation-102lp5a"}, {"title": "Personal Data, Interrupted. Revisiting the notion of personal data in the recent CJEU case law and Digital Omnibus proposal | Timelex", "url": "https://www.timelex.eu/en/blog/personal-data-interrupted-revisiting-notion-personal-data-recent-cjeu-case-law-and-digital"}, {"title": "Breyer case of the Court of Justice of the European Union: IP addresses and the personal data definition", "url": "https://arxiv.org/pdf/2511.01751"}]

Based on the search results, I found relevant information about the CJEU cases you mentioned. Here are the key findings:

Breyer Case - "Means Reasonably Likely" Standard

In Breyer (2016), the CJEU found that dynamic IP addresses may be personal data where a controller has legal means to obtain additional information from another party; identifiability depends on "means reasonably likely" to be used in context. The CJEU explicitly noted the importance of context when assessing the 'means likely reasonably to be used' to identify the data subject.

Nowak Case and Personal Opinions

In Nowak (2017), exam scripts and examiners' comments can be personal data because they communicate opinions about, and of, identifiable persons. The Court confirmed that "the particular nature of personal opinions or views [are] as an expression of a person's thinking... necessarily closely linked to that person"

Relative vs. Absolute Identifiability

The search results reveal a critical distinction regarding pseudonymisation: Identifiability must be assessed from the perspective of each actor, taking into account the means reasonably likely to be used to de-anonymise. The same pseudonymised dataset may constitute personal data for the transferor (who can re-identify it using their key) but not for the recipient (who cannot). This reflects relative anonymisation, as opposed to absolute anonymisation, where re-identification is factually impossible for anyone, including the transferor.

Singling Out

To determine whether a natural person is identifiable, account should be taken of all the means reasonably likely to be used, such as singling out, either by the controller or by another person to identify the natural person directly or indirectly.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

12. assistant (2026-06-16T02:58:45.200Z)

This is the most significant development for my skeptical brief: the **EDPS v SRB (C-413/23 P)** CJEU decision from September 2025 substantially shifted the pseudonymisation/anonymisation analysis toward a *relative* and *context-dependent* standard. This is a double-edged sword for the bottom line ? it could either support "publish coordinates freely" (recipient can't re-identify) OR undermine the rigid "pseudonymized data is always personal data" framing. Let me fetch the authoritative legal commentary on this, since it directly bears on claims #1, #2, and the "publish coordinates" practical advice.

13. user (2026-06-16T02:58:57.412Z)

Web search results for query: "pose estimation skeleton keypoints GDPR personal data Article 9 gait recognition anonymization re-identification regulator guidance"

Links: [{"title": "Privacy Beyond the Face: Assessing Gait Privacy Through Realistic Anonymization in Industrial Monitoring - PMC", "url": "https://pmc.ncbi.nlm.nih.gov/articles/PMC12788357/"}, {"title": "Towards a Deeper Understanding of Skeleton-based Gait Recognition", "url": "https://arxiv.org/pdf/2204.07855"}, {"title": "Multi-Biometric Feature Extraction from Multiple Pose Estimation Algorithms for Cross-View Gait Recognition", "url": "https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11645053/"}, {"title": "2D Gait Skeleton Data Normalization for Quantitative Assessment of Movement Disorders from Freehand Single Camera Video Recordings", "url": "https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9185346/"}, {"title": "Human Pose Estimation: all you need to know - Supervisely", "url": "https://supervisely.com/blog/human-pose-estimation/"}, {"title": "Privacy-Aware Video-Based Methods for Gait and Frailty Recognition in Active and Assisted Living Environments | Springer Nature Link", "url": "https://link.springer.com/chapter/10.1007/978-3-031-84158-3_6"}, {"title": "Human Body Pose Estimation for Gait Identification: A Comprehensive Survey of Datasets and Models | ACM Computing Surveys", "url": "https://dl.acm.org/doi/10.1145/3533384"}]

I've performed the web search for your query on pose estimation skeleton keypoints, GDPR, gait recognition, and anonymization topics. Here are the key findings:

Gait Recognition and Privacy Concerns:

Gait recognition is a behavioral biometric technique that identifies individuals based on their unique walking patterns, enabling long-distance identification. Model-based approaches using skeletal key points offer robustness against covariates, with advances in human pose estimation (HPE) algorithms using convolutional neural networks (CNNs) facilitating the extraction of skeletal key points.

Anonymization and Re-identification Risks:

A critical finding from recent research indicates that anonymization removes appearance but leaves gait identity largely intact, indicating that pose-driven anonymization is insufficient for privacy protection. More specifically, realistic full-body anonymization suppresses appearance but preserves motion-based identity, and pose-preserving anonymization does not break gait identity.

GDPR Compliance Challenges:

Skeleton-based methodologies, although promising in reducing identifiable visual data, still require careful handling of joint data to prevent potential privacy breaches. Strategies such as data obfuscation and the addition of noise to gait patterns have been proposed to mitigate privacy risks, aiming to balance the utility of gait data for health and authentication purposes while safeguarding individual privacy against adversarial attacks and unauthorized access. Ensuring the privacy and security of gait data through comprehensive and innovative solutions is essential to gaining user trust and compliance with data protection regulations like the GDPR.

The search results focus primarily on technical aspects of gait recognition and privacy risks but do not contain specific regulatory guidance on GDPR Article 9 classification or official regulator statements on this topic.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

14. user (2026-06-16T02:58:59.021Z)

CJEU Decision: EDPS v SRB on Pseudonymised Data (C-413/23 P)

Core Holding on Personal Data Status

The CJEU adopted a **relative/context-specific approach** to identifiability. Pseudonymised data's classification depends on the recipient's perspective: "The same pseudonymised dataset may constitute personal data for the transferor (who can re-identify it using their key) but not for the recipient (who cannot)."

The test examines "means reasonably likely to be used to de-anonymise" from each actor's vantage point—not absolute anonymisation, but rather whether re-identification is factually possible for that particular party.

Treatment of Opinions as Personal Data

The court found that statements expressing viewpoints are inherently linked to individuals and "generally will be 'personal data'" within GDPR scope—even when names are removed. This captures customer reviews, employee testimonials, and survey responses.

Practical Implications for Data Sharing

Transparency obligations persist regardless of recipient anonymisation. Controllers must disclose recipients in privacy notices and consent language even when data appears anonymous to the recipient. As stated: "Controllers should anticipate subsequent disclosures of pseudonymised data at the point of collection."

Key compliance steps include:

- Identifying specific or categorical recipients upfront
- Documenting legal bases for transfers
- Considering data subject access request responses about recipients

The ruling creates opportunities for lawful pseudonymised data partnerships while tightening transferor accountability.

15. assistant (2026-06-16T02:59:11.343Z)

I now have the key counter-authorities. Let me do two final targeted checks: (1) confirm the EDPS v SRB ruling is genuinely a **2025 CJEU appeal** judgment (the supporting findings cite the older 2023 General Court framing via WP136, so I want to make sure I'm citing current law correctly), and (2) check whether the DPDP Act's definition of "tracking" exists in the Rules and whether the s.9(3) ban truly has no purpose limitation ? plus the Digital Omnibus proposal that may further shift the EU definition of personal data. Let me also verify the Breyer/relative-identifiability angle against a second authoritative source.

16. user (2026-06-16T02:59:23.115Z)

EDPS v SRB Pseudonymization Ruling Summary

Case Details:

- **Case Name:** European Data Protection Supervisor (EDPS) v Single Resolution Board (SRB)
- **Case Number:** T-413/23P
- **Court:** Court of Justice of the European Union (CJEU)
- **Date:** 4 September 2025
- **Type:** Appeal judgment setting aside the General Court's 2023 decision

Key CJEU Holdings on Pseudonymized Data

****1. Recipient-Specific Identifiability Standard****

The CJEU confirmed that pseudonymized data need not constitute personal data for every recipient. Where effective measures prevent reidentification and the recipient cannot reidentify data subjects through "means reasonably likely to be used," the data remains non-personal for that recipient.

The Court stated: "pseudonymized data are not necessarily personal data for every recipient" and that identification is not reasonably likely where risk is "insignificant" or requires "disproportionate effort."

****2. Critical Caveat?Controller Perspective on Transparency****

The transparency obligation to disclose data recipients arises at collection and is assessed from the **controller's viewpoint**, not the recipient's later capabilities. This means controllers must identify recipients in privacy notices regardless of whether data will later be pseudonymized for sharing.

****3. The "Means Reasonably Likely" Test Remains Context-Specific****

The judgment offers no quantitative thresholds. Organizations must assess "all objective factors, such as the costs of and the amount of time required for identification, taking into consideration the available technology" and revisit assessments as technology evolves.

Important Limitation

The ruling does not settle many pseudonymization versus anonymization questions and provides only "modestly" practical guidance for determining when data genuinely becomes non-personal for third parties.

17. user (2026-06-16T02:59:32.636Z)

Web search results for query: "DPDP Rules 2025 definition of "tracking" "behavioural monitoring" children scope all monitoring or advertising data fiduciary obligations"

Links: [{"title":"DPDP Rules 2025: India's Complete Compliance Guide | Seclore","url":"https://www.seclore.com/fundamentals/dpdp-rules-2025-compliance-guide/"},"{"title":"DPDP Rules - Does Tracking Children Sans Parental Consent Protect Them?","url":"https://www.medianama.com/2025/11/223-dpdp-rules-tracking-children-parental-consent/"},"{"title":"New Data Rules Allow Child Tracking for Safety Without Consent","url":"https://www.medianama.com/2025/11/223-dpdp-rules-real-time-child-tracking-without-consent/"},"{"title":"DPDP Rules 2025 Explained: Full Overview and Practical Summary","url":"https://tsaaro.com/blogs/dpdp-rules-2025-explained-full-overview-and-practical-summary"},"{"title":"DPDP Act 2025: Rules, Compliance Requirements & Penalties Explained","url":"https://www.prophaze.com/dpdp-act-2025-rules-compliance-penalties/"},"{"title":"Rethinking Regulation of Behavioural Monitoring under DPDP Act","url":"https://www.medianama.com/2025/03/223-rethinking-behavioural-monitoring-targeted-ads-regulation-dpdp-act/"},"{"title":"Children's Data Under the DPDP Act, 2023 and DPDP Rules, 2025: The New Compliance Frontier for EdTech, Gaming, Social Media, and Consumer Platforms","url":"https://ksandk.com/data-protection-and-data-privacy/childrens-data-protection-under-indias-dpdp-rules/"},"{"title":"Processing of children's personal data under the DPDP Act: what does it mean for businesses? - ELP Law","url":"https://elplaw.in/leadership/processing-of-childrens-personal-data-under-the-dpdp-act-what-does-it-mean-for-businesses-3/"},"{"title":"Children's Data ? Section 9 of the DPDP Act · DPDP India","url":"https://www.dpdhindia.in/dpdp-children.html"}]

Based on the search results, here's information about the DPDP Rules 2025 regarding tracking, behavioral monitoring, and children's data:

Definition and Scope of "Tracking" and "Behavioral Monitoring"

Data Fiduciaries must NOT undertake tracking or behavioral monitoring of children, which covers cookie-based tracking, device fingerprinting, location tracking, activity monitoring, cross-site tracking, and all other forms of behavioral surveillance of users under 18.

More specifically, data fiduciaries are not allowed to: (i) behavioral monitoring (keeping track of children's online activities, app usage, or browsing patterns that aren't necessary for providing a service), (ii) targeted advertising (using profiling or behavioral data to send personalized commercial messages to children), and (iii) certain types of profiling.

General Prohibitions and Fiduciary Obligations

Data Fiduciaries must not track, monitor, or profile children, or engage in behavioral targeting directed at children. Platforms must not serve targeted advertising to children.

Scope - Advertising Data and Monitoring

Section 9 of the Act prohibits platforms from tracking or carrying out behavioral monitoring of children or targeted advertising directed at children, unless exempted. The rules do permit limited exceptions—for example, service providers are exempt from obtaining verifiable parental consent and from the restrictions on tracking and behavioral monitoring for determining the real-time location of a child, when limited to real-time location tracking in the interest of a child's safety and protection.

Age Threshold

The DPDP Act defines a "child" under Section 2(f) as any individual below the age of 18 years.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

18. assistant (2026-06-16T03:00:08.113Z)

I have enough authoritative material to render a skeptical verdict. Let me note the one citation correction I found (the case number is C-413/23 P at the CJEU, the Clifford Chance page labels it "T-413/23P" but it's the appeal judgment of 4 September 2025 setting aside the General Court ? I'll flag the verified date/court and not over-claim the exact docket format).

My skeptical analysis identifies several material qualifications and one near-refutation of an over-confident reading. Here is my brief.

****The bottom line is directionally sound but contains at least four over-confident statements that could get the client in trouble, plus it omits a 2025 CJEU development that cuts BOTH ways.****

1. *****"Shuttle tracks = publish freely" is too absolute.**** The findings themselves admit the WP136 content/purpose/result test and that "data relating to an object can identify a person." For a ***named match*** (which is the entire point of a highlight indexer ? the footage is of identifiable players in a known event), shuttle trajectories + timestamp + match identity become ***relatable*** to the players via the "purpose/result" limb and Recital 26 "singling out." "Publish freely" is safe only for genuinely context-free coordinates; tied to a published reel of an identifiable match, even shuttle data sits inside a personal-data processing operation. Qualify to: shuttle coordinates are not personal data ***in isolation***, but publishing them as part of identifiable match footage does not launder the surrounding personal-data obligations.

2. ****The "pseudonymized data is always personal data" framing is now OUTDATED ? refuted by CJEU.**** The supporting JSON relies on the old absolute reading of Recital 26. The CJEU appeal judgment ****EDPS v SRB, 4 September 2025**** (Court of Justice, appeal setting aside the General Court's 2023 ruling) held that ****pseudonymised data is NOT necessarily personal data for every recipient**** ? identifiability is ***relative and recipient-specific***, turning on whether re-identification is "reasonably likely" / not requiring "disproportionate effort" ***for that actor***. This is a double-edged sword: (a) it ***helps*** the client (coordinates a third party genuinely can't relink may be non-personal in that recipient's hands), but (b) the

****transferor/controller still treats the data as personal**, and transparency/recipient-disclosure duties persist *from the controller's viewpoint at collection*. An over-confident lawyer citing only the old "pseudonymised = always personal" rule would be citing superseded authority. Sources: Clifford Chance, Osborne Clarke.**

3. *****s.9(3) flatly BANS tracking/behavioural monitoring" ? materially over-stated; ignore Rule 12 / Fourth Schedule and the safety carve-out at your peril.**** The DPDP Rules 2025 (Rule 12 + Fourth Schedule Parts A/B) create real exemptions from BOTH s.9(1) verifiable consent AND the s.9(3) ban for specified classes (healthcare, schools, day-care, child-transport) and purposes (e.g., ****real-time location tracking in the interest of a child's safety****). Two caveats that protect the client from over-reliance: (i) an amateur badminton highlight indexer is ****not**** within those exempt classes, so the exemptions almost certainly ****don't rescue**** this use case ? the ban still bites; (ii) but the bottom line's "flatly BANS, no exceptions" phrasing is legally inaccurate as a statement of the law. More importantly, the commentary (medianama, cyberpeace) flags that "tracking/behavioural monitoring" is ****undefined**** in the Act and its breadth is genuinely contested ? so the claim that "a persistent pose/gait signature **arguably is**" behavioural monitoring is defensible but ****uncertain****, not settled. The honest posture: the s.9(3) risk for a persistent per-player movement signature is **real* and the conservative reading*, but it rests on an undefined term, not black-letter certainty.

4. ****The Art.9 line ("not biometric as long as the purpose is detect/categorize") is correct but fragile in this product's context.**** EDPB Guidelines 3/2019 do support that categorization-without-unique-identification escapes Art.9. BUT the indexer's **whole value proposition** trends toward per-player tracking ("named player + gait/pose signature"). The moment pose tracks are used to ****distinguish/follow a specific player across rallies or videos**** (re-identification), the purpose flips to "uniquely identifying," and Art.9 attaches. So "bbox/pose coordinates detached from identity = generally OK" is true **only** while the system genuinely does not single out a specific individual ? a condition this product is architecturally prone to violate. The gait-anonymization research (Weiß et al. 2025) is solid and correctly cited; it strengthens the "pose tracks are not reliably anonymous" point.

****Net verdict: QUALIFIED.**** The triage framework and the practical publish/no-publish ladder are sound engineering-grade guidance and align with the project's existing guardrails (exclude minors from training, gait/biometrics strictly future, generic schema). But four statements are over-confident in ways that could mislead: "publish freely," the obsolete absolute-pseudonymisation rule, "flatly BANS," and the implicit assurance that pose coordinates stay outside Art.9. The corrected reading is more cautious on shuttle/object data tied to identifiable matches, acknowledges the 2025 CJEU relativity shift, and treats the s.9(3)/gait conclusions as the **conservative** reading of **undefined/contested** terms rather than as certainties.

Let me also note: I was unable to fetch the primary DPDP Act statutory text and the medianama scope article returned 403; those specific claims rest on secondary commentary, so I've kept the s.9(3)-scope conclusions at moderate confidence rather than high.

19. user (2026-06-16T03:00:36.833Z)

Structured output provided successfully

20. assistant (2026-06-16T03:00:47.658Z)

I've completed my skeptical second-lawyer review and returned the structured verdict.

Verdict: ****qualified.**** The triage framework is directionally sound and aligns with the project's guardrails, but I found four over-confident statements and one omitted 2025 development that could mislead an over-confident reader:

1. *****Shuttle tracks = publish freely***** is too absolute ? coordinates tied to an identifiable named match sit inside a personal-data processing operation via the WP136 purpose/result limb and Recital 26 singling-out.
2. ****The "pseudonymised = always personal data" framing is superseded**** by the CJEU appeal judgment in **EDPS v SRB** (4 September 2025), which made identifiability relative/recipient-specific. This cuts both ways: it can help (detached coordinates a recipient can't relink) but the controller still owes transparency/recipient-disclosure duties.
3. *****s.9(3) flatly BANS***** overstates the law ? DPDP Rules 2025 Rule 12 + Fourth Schedule carve

out classes/purposes (including child-safety location tracking). Those don't rescue an amateur highlight indexer, so the ban still bites this client, but the "no exceptions" phrasing is inaccurate, and "tracking/behavioural monitoring" is undefined, so treating a persistent pose/gait signature as prohibited monitoring is the *conservative* reading, not a certainty.

4. **The Art.9 "detect/categorize is fine" point is fragile here** ? the product trends toward per-player tracking, and the moment pose tracks single out a specific player across rallies the purpose flips to unique identification and Art.9 attaches.

Honesty caveats flagged: I could not fetch the primary DPDP statutory text, and the medianama scope article returned 403, so the s.9(3)-scope and gait-as-monitoring conclusions rest on secondary commentary and are held at moderate (not high) confidence. The *EDPS v SRB* case number appears as "T-413/23P" on the Clifford Chance page but is the CJEU appeal judgment of 4 September 2025 setting aside the General Court ? I verified court/date/disposition rather than over-claiming the exact docket format.